# Victorine Younang, Ph.D.

**Research Scientist — Ph.D., Computer Science**

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## SUMMARY

Research Scientist (Ph.D., Computer Science) specializing in applied machine learning for cybersecurity, graph representation learning, and LLM alignment. Designs novel architectures for complex risk modeling and vulnerability detection, combining probabilistic graphical models (Bayesian Attack Graphs) with deep learning (GNNs, MoE LLMs). Strong publication record across top-tier security, AI, and systems venues (IEEE TDSC, IEEE TSC, NDSS, AAAI); bridges theoretical ML with empirical evaluation on real-world, large-scale data.

## TECHNICAL SKILLS

- **Research & Modeling:** deep-learning architectures, Graph Neural Networks (GAT/ST-GNNs), probabilistic graphical models (Bayesian inference, Markovian analysis), causal inference, representation engineering, physics-informed ML, ablation studies
- **LLMs & Generative AI:** foundation-model fine-tuning (LoRA/QLoRA), Mixture-of-Experts (MoE), Direct Preference Optimization (DPO), curriculum learning, automated evaluation-metric design, RAG
- **Core ML / NLP:** PyTorch, PyTorch Geometric, Hugging Face, Transformers (BERT/BioBERT), Node2Vec, Vector AutoRegression (VAR), anomaly detection in highly imbalanced regimes
- **Experimentation & Systems:** Python, R, Distributed Data Parallel (DDP), gradient checkpointing, MLflow, 4-bit quantization, HPC cluster execution, Hadoop/HDFS

## EXPERIENCE

### Oakland University | Machine Learning Researcher

*Michigan, USA · Sep 2020 – Apr 2026*

- **Curriculum Learning & LLM Benchmarking (TIA):** orchestrated curriculum fine-tuning of six foundation LLMs (incl. DeepSeek-V2 MoE, Qwen2.5) to evaluate inductive biases across dense and MoE architectures; designed three neural-symbolic evaluation metrics (syntactic validity, semantic match, value consistency) measuring true generative correctness beyond textual overlap.
- **Graph Representation Learning (CryptoGraph):** engineered a hybrid Graph Attention Network + MLP in PyTorch Geometric to learn structural representations from complex AST graphs; solved small-data, severe class-imbalance regimes via variant-aware stratified sampling and GraphNorm → robust zero-shot generalization (F1 0.94).
- **Probabilistic Risk Modeling:** developed a complex-valued Bayesian Attack Graph modeling concurrent adversarial interactions and cyclic threat dependencies for real-time risk scoring; published in IEEE TDSC (13-point risk-delta improvement over static models on a connected-vehicle network).

### Health Alliance Plan (HAP) | ML Researcher (NLP & Multimodal AI)

*Detroit, MI · Summer 2025*

- **Multimodal Representation Learning:** researched and benchmarked multimodal deep-learning systems fusing BioBERT/ClinicalBERT text embeddings with structured tabular features to predict high-risk 30-day hospital readmissions.
- **Algorithmic Fairness & Interpretability:** conducted rigorous bias audits across demographic subsets (age, zip) and applied glass-box interpretability (SHAP, LIME) to translate deep classifier embeddings into transparent risk explanations.

### Rec Room | Applied ML Research Intern (Algorithms)

*Seattle, WA · Summer 2023*

- **Algorithmic Recommender Systems:** researched and developed a structural content-recommendation algorithm (Node2Vec, Random Walk embeddings) across a bipartite graph of 1,000+ virtual environments.
- **Empirical Validation & Cold-Start Logic:** designed cold-start logic from structural graph properties; validated via rigorous online A/B testing → statistically significant +33% average session time.

### EDF (Électricité de France) | R&D ML Researcher (Graph ML & Time-Series)

*Paris, France · Summer 2021*

- **Physics-Informed Deep Learning:** researched and trained deep Spatial-Temporal GNNs (ST-GNNs) in PyTorch capturing non-linear temporal and physical spatial dependencies across low-voltage grids.
- **Probabilistic Loss Formulation:** engineered probabilistic forecasting architectures (quantile/Pinball loss, Gaussian processes) generating strict 24-hour confidence intervals.

- **Custom Regularization:** developed loss regularizers penalizing physically impossible predictions so deep sequence models natively respected hardware capacity constraints.

**EDF (Électricité de France) | Big Data & Machine Learning Engineer**
*Paris, France · 2019–2020*

- **Large-Scale Data Architecture for ML:** managed the big-data architecture for a consent platform on distributed Hadoop clusters, establishing the high-volume data foundation for downstream analytics and research.
- **Algorithmic Data Validation:** researched and engineered GDPR data-sanitization and validation workflows, eliminating data contamination and reducing downstream ML training errors by 30%.

**Orange Telecom | Applied ML Researcher**
*Paris, France & Brussels, Belgium · 2017–2019*

- **Time-Series Forecasting Models:** developed multivariate predictive models (VAR, ARIMA) in R forecasting distributed Hadoop connection volumes and query loads → proactive cluster compute allocation.
- **High-Dimensional Data Parsing:** engineered high-throughput Python ingestion pipelines processing complex, unstructured XML network logs into optimized vector/feature stores for real-time telemetry analysis.

## SELECTED PROJECTS

- **CipherGuard — reference-free backdoor detection** (GAT + AST, Isabelle/HOL-verified corpus): catches test-vector-compliant backdoors with zero false positives on 44 ciphers (NDSS 2027).
- **RepoSage — Agentic, execution-verified RAG** (LangGraph, vLLM, Qdrant, DPO): dual-source code-&-docs RAG that generates, executes in a sandbox, and self-corrects against a library's own tests; DPO alignment from execution-feedback preferences.
- **LLM Deception & Honesty Probing:** representation-engineering probes across Qwen3 (0.6B–32B); found the probe keyed on persona/chat-template tokens rather than content, and proposed incentive-based training.

## EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (Sep 2020 – Apr 2026; GPA 3.96). Dissertation: *From Perceived Trust to Provable Security: Design, Optimization & Verification of Resource-Constrained IoT Ecosystems.*
- **M.Sc. Computer Engineering & B.Sc. Electronic & Computer Engineering** — ESIGELEC, France (2016–2019; GPA 3.87). *Big Data for Digital Transformation.*

## SELECTED PUBLICATIONS

- V. Younang et al. "Security Risk Assessment with Bayesian Attack Graphs and Complex Probabilities." *IEEE TDSC*, 2025.
- V. Younang et al. "Extending IoT Devices' Network Lifetime." *IEEE TSC*, 2025.
- V. Younang et al. "Comparative Analysis of Users' Concerns and Government Policies on Autonomous Vehicles." *Springer ICIOT / LNCS*, 2022.
- V. Younang et al. "CipherGuard: Reference-Free Detection of Backdoors in Cipher Implementations." *Under review, NDSS 2027.*
- V. Younang et al. "Tiered Isomorphic Alignment: Curriculum Fine-Tuning across Six LLMs." *Under review, AAAI 2027.*

## CERTIFICATIONS

- **AWS Certified AI Practitioner (AIF-C01)** — expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — expected Dec 2026; transformers, prompt engineering, fine-tuning, NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated in RepoSage.